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Average

First 'n' odd numbers = n

First 'n' even numbers = $n+1$

First 'n' natural numbers = $\frac{n+1}{2}$

Sum = $n * n = n^2$

Sum = $(n+1) * n = n(n+1)$

Sum = $\frac{n+1}{2} * n = \frac{n(n+1)}{2}$

$Avg = \frac{Sum}{n}$

$Sum = Avg * n$

Share

Average

9:42:34 / 11:50:56

TCS Average

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Q4. The average of first 30 odd numbers. n 30

(a) 30 ✓ (b) 60 (c) 90 (d) 120

Q5. The average of first 30 even numbers. $n+1$ $30+1=31$

(a) 31 ✓ (b) 61 (c) 91 (d) 121

Q6. The average of first 30 natural numbers. $\frac{n+1}{2}$ $\frac{30+1}{2} = \frac{31}{2} = 15.5$

(a) 15.5 ✓ (b) 14.5 (c) 16.5 (d) 17.5

Average 9:43:09 / 11:50:56 TCS Average >

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Q8. The average salary of all the employees in a small organization is Rs 8,000. The average salary of 7 technicians is Rs 12,000 and the average salary of the rest is Rs 6,000. The total number of employees in the organization is ?

A. 21

B. 20

C. 26

D. 22

$$8000 = \frac{(12000 \times 7) + 6000x}{7+x}$$

$$\overset{4}{\cancel{8}} = \frac{(\overset{6}{\cancel{12}} \times 7) + \overset{3}{\cancel{6}}x}{7+x}$$

$$4 = \frac{(6 \times 7) + 3x}{7+x}$$

$$4(7+x) = 42 + 3x$$

$$28 + 4x = 42 + 3x$$

$$4x - 3x = 42 - 28$$

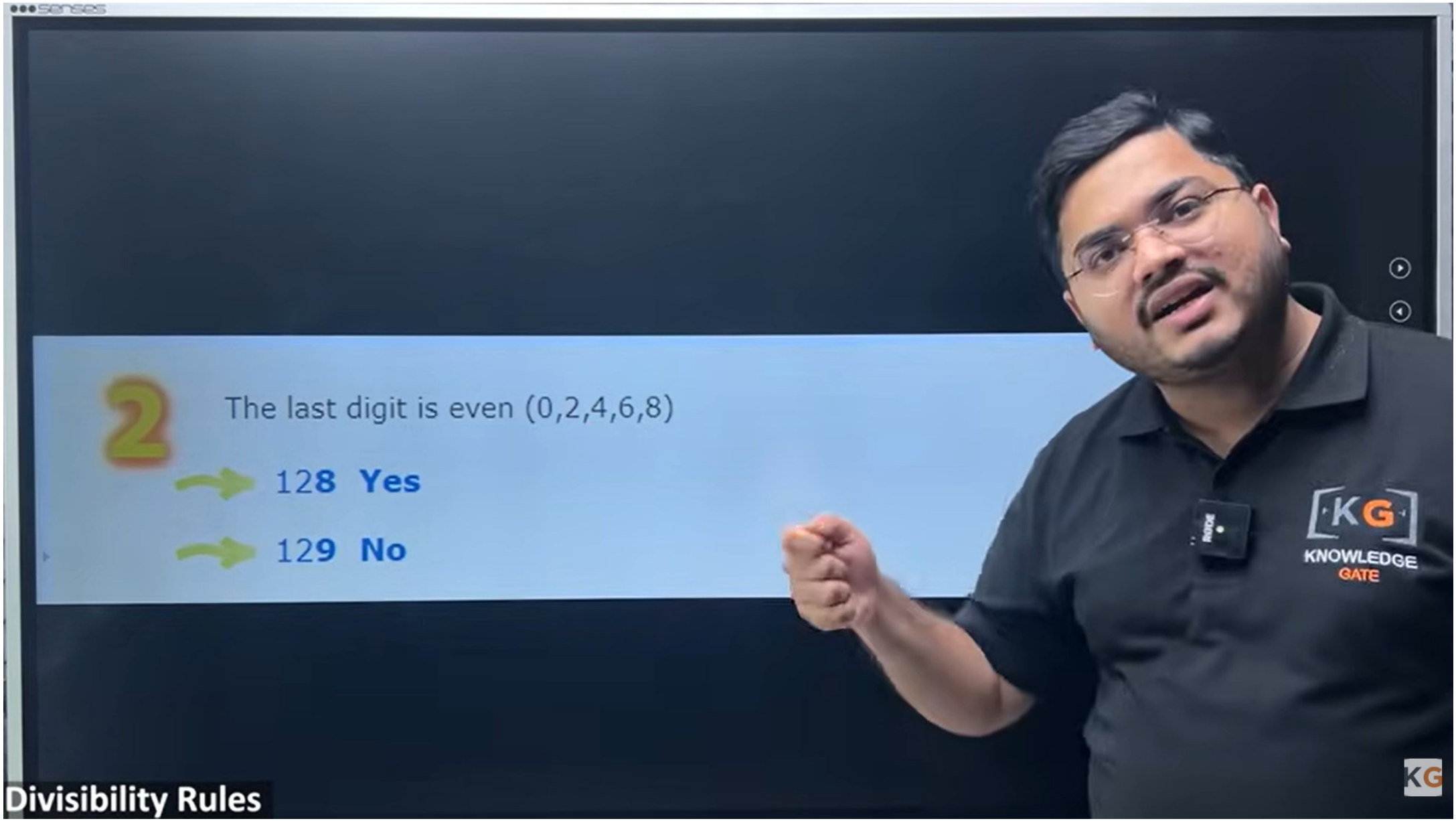
$$x = 14$$

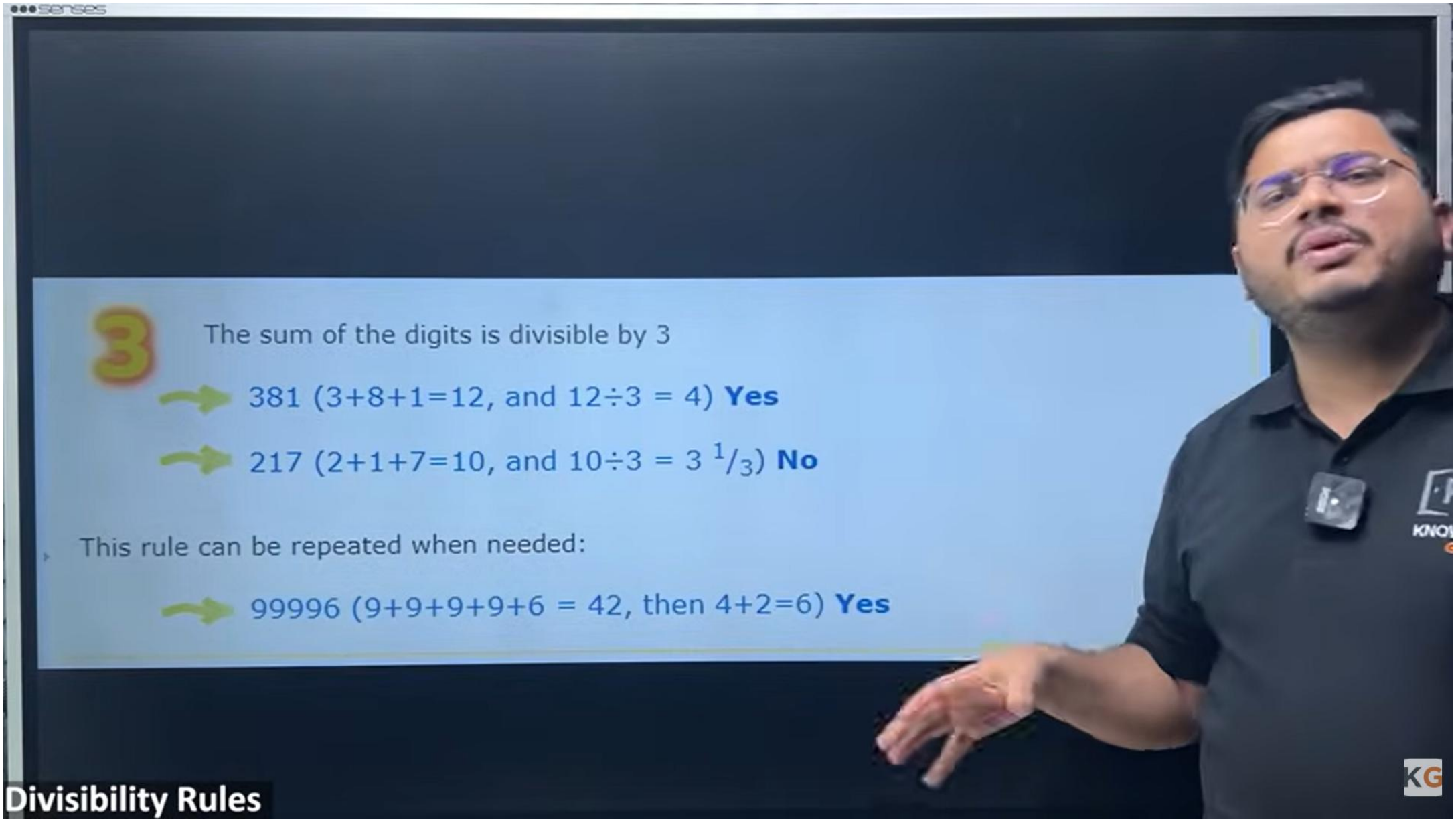
Total Employees $= 7 + x$

$= 7 + 14 = 21$

TATA CONSULTANCY SERVICES

9:48:04 / 11:50:56 TCS Average


2
The last digit is even (0,2,4,6,8)
128 Yes
129 No
KG
KNOWLEDGE
GATE
Divisibility Rules


3
The sum of the digits is divisible by 3
381 (3+8+1=12, and 12÷3 = 4) Yes
217 (2+1+7=10, and 10÷3 = 3 1/3) No
This rule can be repeated when needed:
99996 (9+9+9+9+6 = 42, then 4+2=6) Yes
Divisibility Rules

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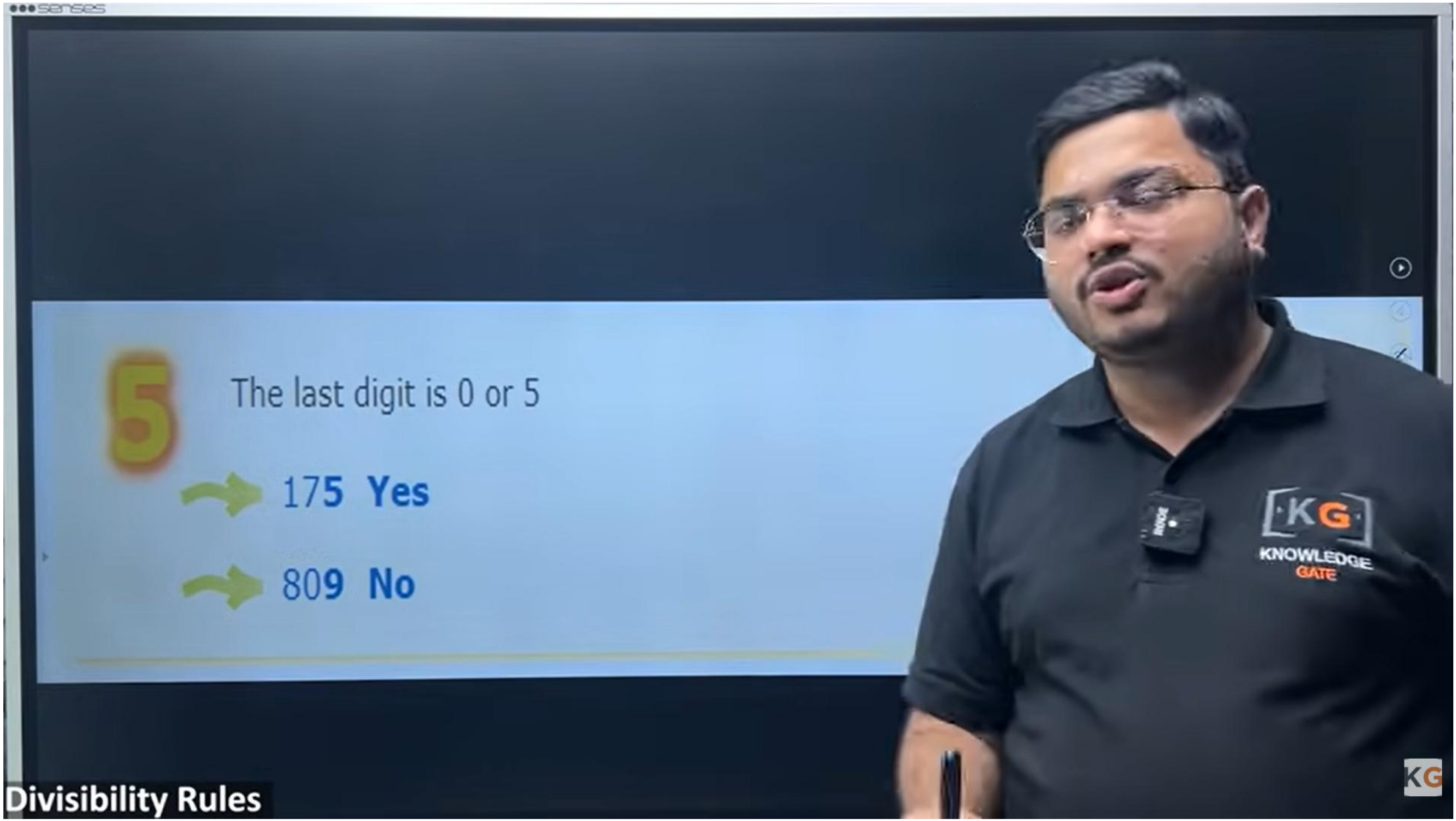
4 The last 2 digits are divisible by 4

- 13**12** is (12÷4=3) **Yes**
- 70**19** is not (19÷4=4 $^{3}/_{4}$) **No**

A quick check (useful for small numbers) is to halve the number twice and the result is still a whole number.

- 1[illegible]2 = 6, 6/2 = 3, 3 is a whole number. **Yes**
- [illegible]2 = 15, 15/2 = 7.5 which is not a whole number. **No**

9:50:47 / 11:50:56 TCS Divisibility Rules
Divisibility Rules


5
The last digit is 0 or 5
175 Yes
809 No
KG
KNOWLEDGE
GATE
Divisibility Rules

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$6 = \{1, 2, 3, 6\}$

$\frac{N}{6}$

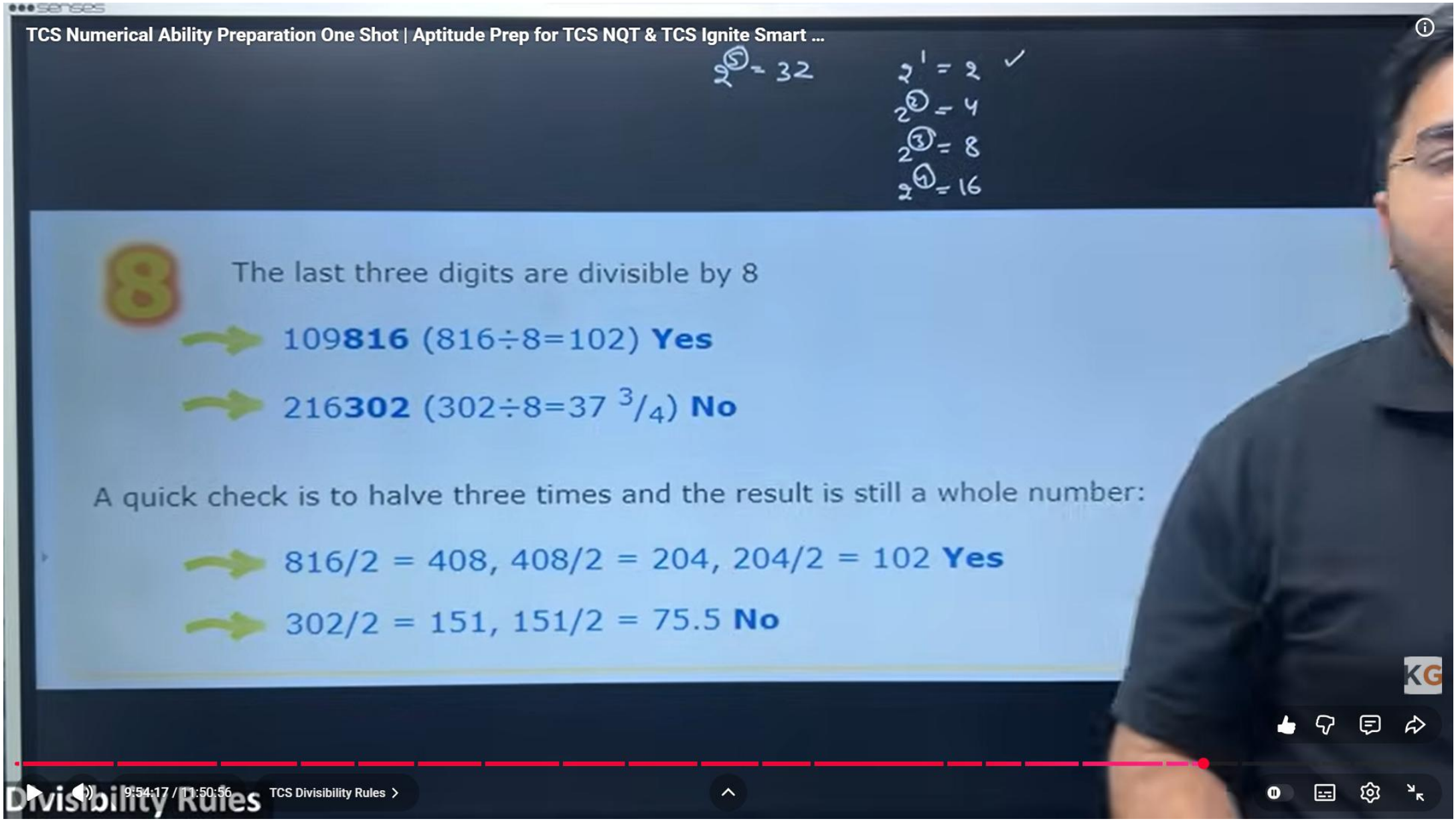
$\frac{N}{2}$ $\frac{N}{3}$

6 Is even and is divisible by 3 (it passes both the 2 rule and 3 rule above)

- 114 (it is even, and 1+1+4=6 and 6÷3 = 2) **Yes**
- 308 (it is even, but 3+0+8=11 and 11÷3 = $3\,^{2}/_{3}$) **No**

Divisibility Rules

9:52:31 / 11:50:56 TCS Divisibility Rules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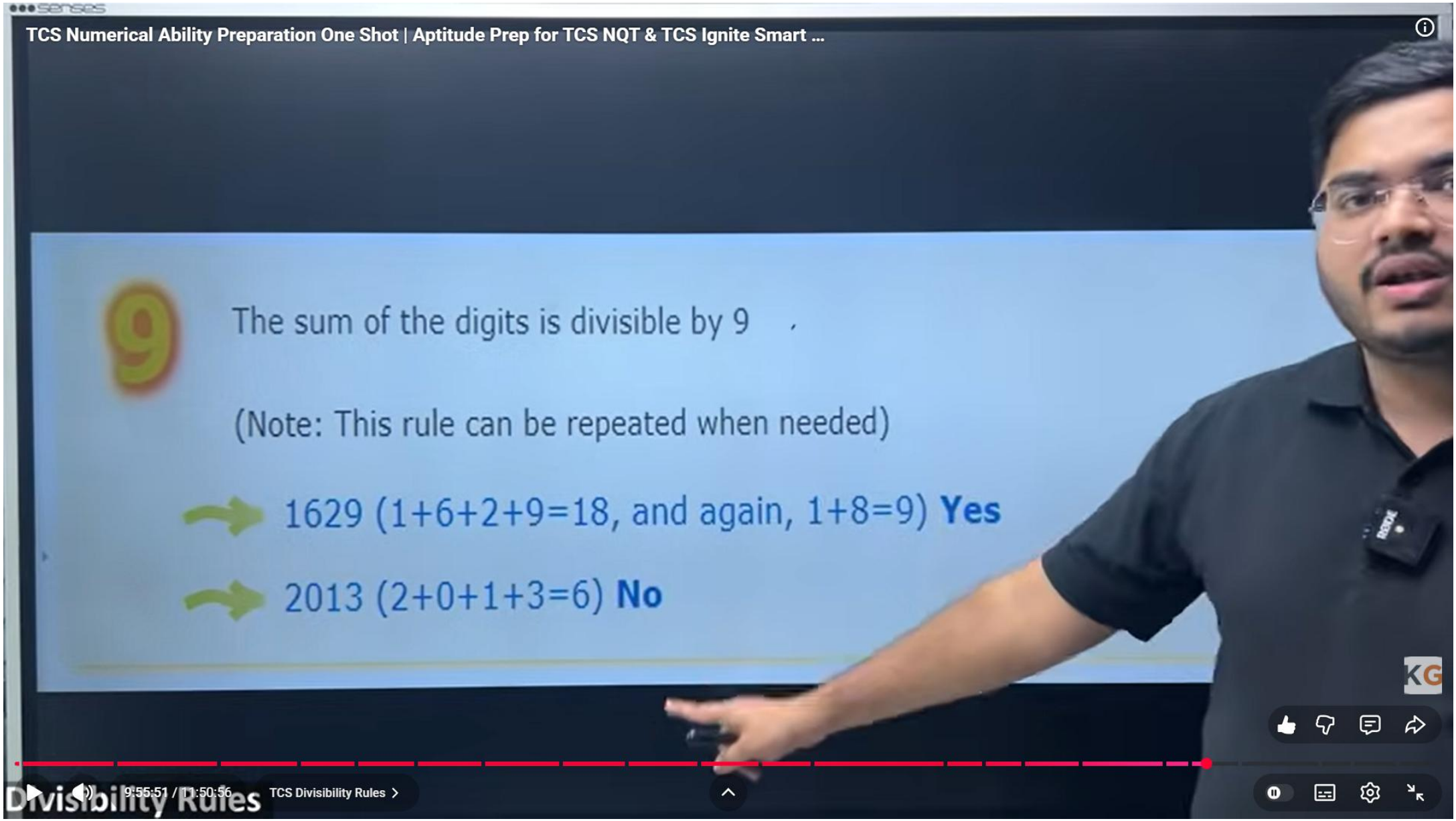
$2^5 = 32$
$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16$
8
The last three digits are divisible by 8
109816 (816÷8=102) Yes
216302 (302÷8=37 3/4) No
A quick check is to halve three times and the result is still a whole number:
816/2 = 408, 408/2 = 204, 204/2 = 102 Yes
302/2 = 151, 151/2 = 75.5 No
Divisibility Rules
9:54:17 / 11:50:56
TCS Divisibility Rules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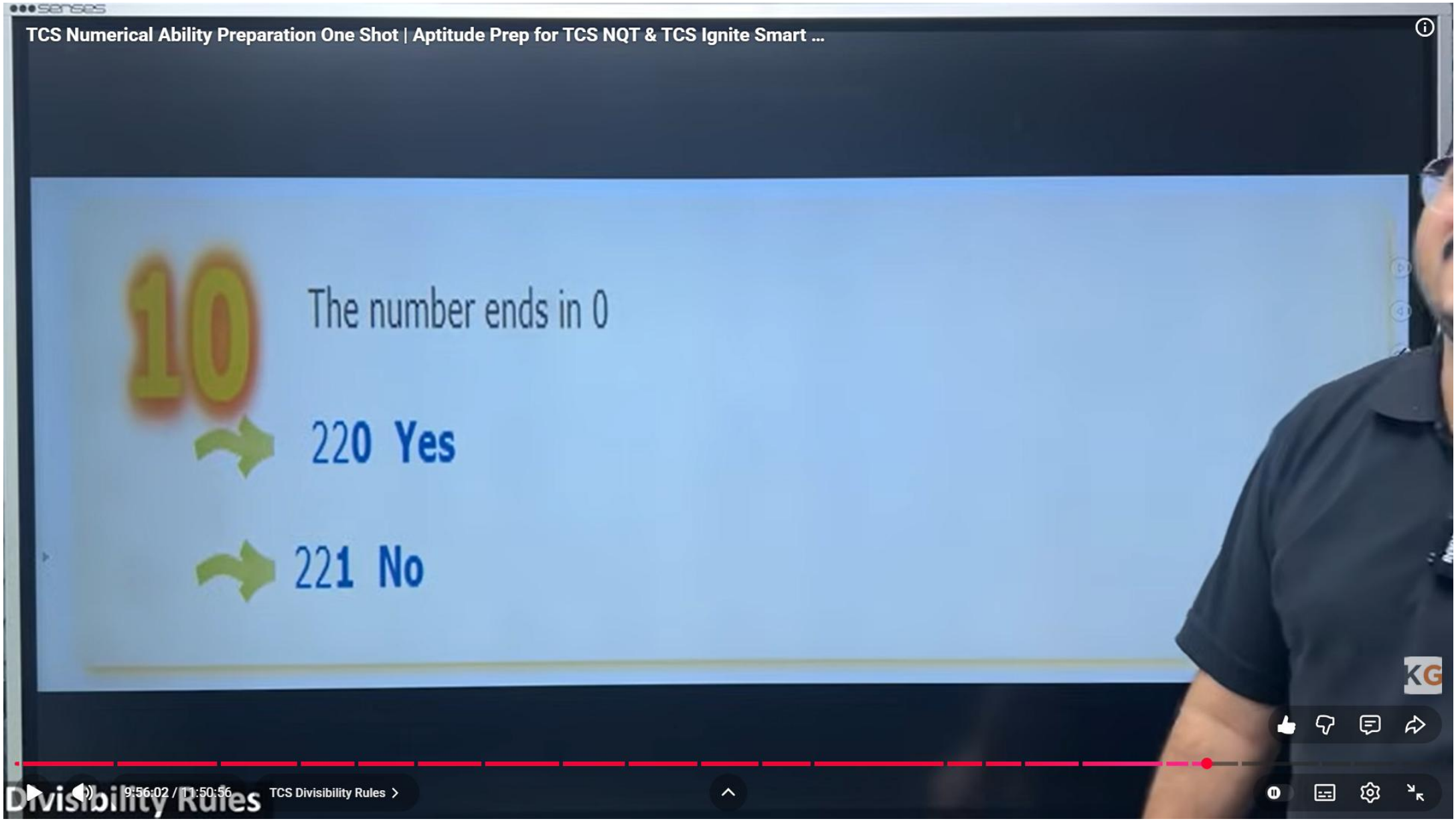
9
The sum of the digits is divisible by 9
(Note: This rule can be repeated when needed)
1629 (1+6+2+9=18, and again, 1+8=9) Yes
2013 (2+0+1+3=6) No
Divisibility Rules
9:55:51 / 11:50:56
TCS Divisibility Rules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10
The number ends in 0
220 Yes
221 No
Divisibility Rules
9:56:02 / 11:50:56
TCS Divisibility Rules

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11

Add and subtract digits in an alternating pattern (add digit, subtract next digit, add next digit, etc). Then check if that answer is divisible by 11.

- 1364 (+1−3+6−4 = **0**) **Yes**
- 913 (+9−1+3 = **11**) **Yes**
- 3729 (+3−7+2−9 = **−11**) **Yes**
- 987 (+9−8+7 = **8**) **No**

1+6=7
3+4=7

7−7=0 / 11n ✓

5−16=−11

Divisibility Rules
9:57:52 / 11:50:56
TCS Divisibility Rules